Modelling 3 Final Project Group 12

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Foreword

This project will be about statistically analysing if robot-assisted learning has a positive influence on patients with different personality traits playing a game to help regain control of their hands with the help of a dataset. This project was written for the subject TW2-21 Modelling 3. With the help of prior acquired knowledge about statistics and the programming language R, the data provided will be analyzed using statistical methods and R-functions to then come to certain conclusions.

Furthermore, an extension was added to further analyze the data and to get a clearer picture on how the dynamics of robot-assisted learning together with certain personality traits function. This project, together with its extension, might provide valuable insight on how robot-assisted learning can aid patients trying to complete a task. The focus of this project is placed on helping injured patients play a game to regain control of their hands, but in a hospital for example, it might be valuable in helping patients with a bigger or other physical injury to rehabilitate.

To get a clearer understanding of the report, it is recommended that the reader has a bit of knowledge about statistics and how it can be used to analyze data.

As last, we would like to thank Dr. Ir. Tina Nane for her knowledge, advice and support during the making of this project. [1]. For the use of generative AI see the footnote.

Delft, January 20, 2026

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[1] Generative AI was used to enhance grammar and spelling of the report aswell as debugging the code and making it efficient

Summary

This project investigated whether or not robot-assisted learning reduced errors in a simple target game across one hundred injured patients with different personality profiles. The patients completed a personality questionnaire measuring four personality traits labeled: AC, FS, TC and TB and then played the game in three stages: baseline (BL), short term retention (STR), and long term retention (LTR). Analyses combined descriptive plots, distribution fitting, hypothesis tests and linear regression to link personality, training and robot guidance to performance measured by average absolute error.

Personality scores were modeled with standard continuos families. FS and TC were best described with beta distributions, AC by a Weibull distribution and TB by a normal distribution. fit tests and QQ plots indicated that these choices were reasonable approximations of the score distributions. Initial regression results show FS raises baseline error while TC is associated with lower error at baseline.

Preliminary results indicate training strongly reduces error between BL and both STR and LTR. Comparison of control and experimental groups finds no significant group difference at BL but shows significant differences at STR and LTR. Extended regression models with group interactions reveal that robot guidance interacts with personality: guidance yields measurable error reductions for certain trait profiles after training, notably for FS and later for AC.

Contents

1 Introduction

After having experienced traumatic events like stroke or having been severely injured, individuals often need rehabilitation in order to recover from their injuries. This can be quite an intense and long process and the duration of rehabilitation may even go up to 7 to 8 hours a day, 7 days a week, which was the case for patients who had to recover from brain injury (Blackerby, 1990). To help shorten this process and to support medical care, modern technologies, including robotics, are being developed in order to aid these goals. This modernisation using robotics has already been proven to be effective in some cases when it comes to traumatic injury, for example in the restoration of functionality to a partially or fully paralysed upper or lower limbs (Bogue, 2018). Since it has already been proven that robotics can sometimes be useful in the process of rehabilitation, the goal of this research is to answer the following research question: **"Can robot-assisted learning help patients with different personality traits reduce errors in a game?"**. The answer to this research question may provide useful insights on how robot-assisted learning helps patients with a wide range of personality traits complete a simple task that mimics real-life scenarios, and these insights may be used in rehabilitation.

To analyse if robot-assisted learning has an effect of error reduction on the patients, the patients get divided into two groups and three categories. The groups consist of the "Control"-group (C) and "Experimental-group". The patients for the control-group does not receive any guidance from the robot while the patients for the experimental group do receive guidance. Patients who are chosen to be in the experimental group are randomly picked out of the 100 patients. The categories are the following: BL, STR and LTR. The patients play the game once to get used to the controls and the game itself, at this point the scores are not logged yet. After that the patients play the game twice, these scores are logged and fall under the stage BL. After the patients played the game twice, they take a short break of 10 minutes, after the break they will play the game twice again, this time the scores are logged too and these fall under the stage STR. After a few days, the patients play the game twice again, these scores are logged again and fall under the stage LTR.

Section 2 will be about analysing personality traits and the data using visual and numerical methods. Section 3 will be about analysing the potential effects of robot-assisted learning using numerical analysis and linear regression. Section 4 will be about validating the conclusions made in section 3 by performing a power analysis. Section 5 will be about analysing dependence between personality traits. The code used for this project can be in the Appendix.

2 Analysis of personality traits

In this section the personality traits of the patients will be analyzed by using an appropriate dataset. In section 2.1 each personality trait will be explained and the frequencies of the scores for each personality will be displayed and connected to their respective personality trait. In section 2.2 the scores displayed will be modeled as probabilistic distributions, and a goodness-of-fit test will be performed in order to test the correctness of the chosen distribution for each personality trait.

2.1 Analysis of personality traits and data

For this project, 100 patients were gathered to play a game where their goal was to minimize the Error each time. Each person has their own personality trait: achiever (AC), freedom of spirit (FS), transformation of challenge (TC) and transformation of boredom (TB). Someone with the achiever personality generally has a preference for getting high scores, someone with the free spirit personality generally has a preference for exploration, someone with the transformation of challenge personality generally has a tendency to transform "challenging" situations into personally motivating ones, and lastly someone with the transformation of boredom personality generally has a tendency to transform "boring" situations into personally motivating ones. Each personality trait will be referred to by their abbreviations. For each trait, each patient had to respond to a questionnaire and they would receive a score from 0-100 based on how strongly they represented a personality trait. Below four histograms illustrating the score distribution for all personality traits are given:

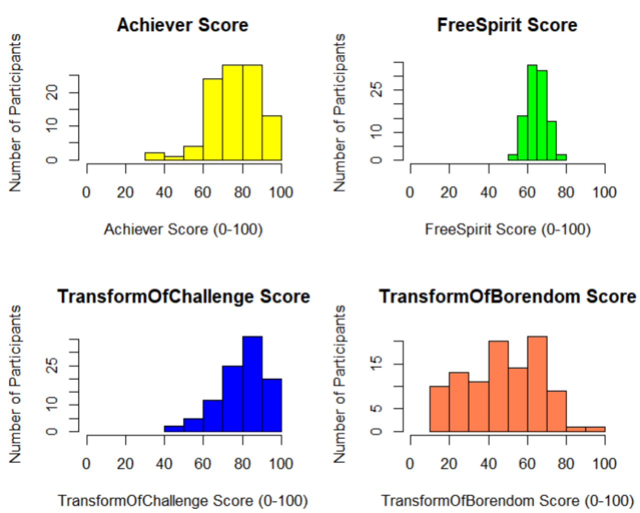

Figure 1: Histograms of personality trait scores

For the personality traits AC and TC it can be seen from the histograms that no patients at all scored 30 or less on their questionnaires. For FS this is even the case for a score of 60 or less, but for this personality trait, there is no-one that scored 80 or higher. This suggests that patients who are in general quite ambitious or like new and/or challenging situations are on average more likely to exhibit their trait, in contrast to patients from the TB personality trait, for whom it is true that the scores are more across the board and that almost nobody scored above 80.

2.2 Fitting and verifying distributions for each personality trait

As seen in section 2.1, the scores for the questionnaires for all four personality traits could be plotted and analysed using histograms. The next logical step would be to try to link the score distribution for each personality trait to a known continuous mathematical distribution and to further analyse that distribution. This step is useful to analyse theoretical properties of the score distribution for each personality trait, for example expectation and variance.

For this project, the following known continuous probability distributions were tested to try and model the score distribution for each personality trait: Exponential distribution, gamma distribution, beta distribution, Weibull distribution and normal distribution. To test which distribution fit each personality trait the best, the Akaike information criterion (AIC) was used to compare the distributions, the statistical metric used to determine the best fit is as follows:

$\min\{\mathrm{AIC}_{\mathrm{exp}}, \mathrm{AIC}_{\mathrm{normal}}, ..., \mathrm{AIC}_{\mathrm{Weibull}}, \mathrm{AIC}_{\mathrm{Beta}}\}$. (1)

Where the Akaike information criterion is defined as follows:

$\mathrm{AIC}_{\mathrm{d}} = 2K_d - 2\log(L_d)$ (2)

Where K_d is the amount of parameters the distribution d takes and where $\log(L)$ denotes the log-likelihood of the chosen distribution d.

The more parameters a distribution has, the bigger the likelihood and thus the log-likelihood of the distribution will be. The AIC-metric prevents this type of overfitting using too many parameters, while also rewarding goodness-of-fit of the chosen distribution.

By calculating the minimimum of the AIC-metrics for all personality traits, adjusting the data to the support of the theoretical distribution (For example for beta which has support $[0, 1]$) and applying a goodness-of-fit test to each distribution, in this case the Kolmogorov-Smirnov (KS)- test, the following results were obtained:

Personality trait	Optimal distribution	Parameters	p-value
FS	Beta	(2.2848, 2.6996)	0.8461
AC	Weibull	(6.9713, 81.9208)	0.8308
TC	Beta	(2.8669, 1.4388)	0.7436
TB	Normal	(47.7600, 19.4854)	0.6847

Table 1: Optimal distribution fits for each personality trait

For each distribution, each p-value is significantly larger than 0.05, which means that there is no statistically significant evidence against H_0 which supports the chosen distributions.

All distributions that were listed above also make "intuitive" sense according to the histograms shown in section 2.1. Personality traits AC, TC and FS received skewed distributions like the beta and Weibull distributions, while TB received a distribution that is not skewed at all, namely the normal distribution, which are good estimates based on the histograms for the personality traits. To illustrate the correctness of the distributions compared to the data for each personality trait, QQ-plots were made to compare and plot the theoretical quantiles against the actual quantiles of the data, the QQ-plots are given below:

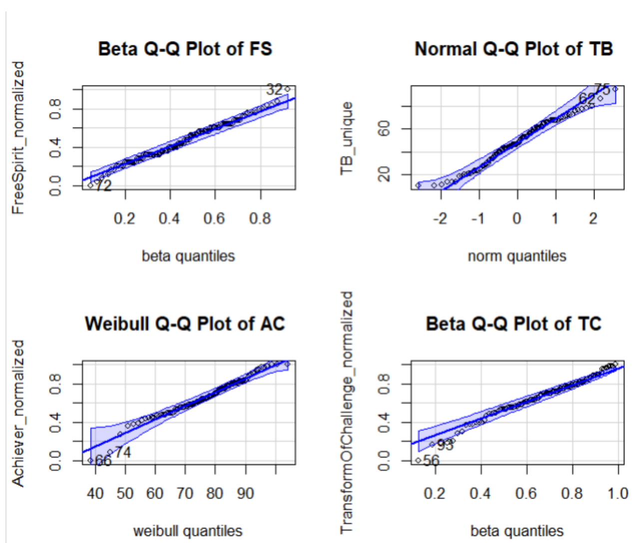

Figure 2: QQ-plots of distributions against data quantiles

It can be seen that most of the data points in the QQ-plots for all personality traits lie on a straight line with the theoretical quantiles plotted against the actual data quantiles. There are also some outliers at the low and high end of the data for all QQ-plots, this signifies that the distributions chosen serve as good approximations for the score distributions of the personality traits, but it also highlights that the distributions chosen do not perfectly capture all the real-life statistical noise in the data. The bands in each figure signifies the QQ confidence bands, which represent the uncertainty of the plots with noisy data.

3 analysing potential effects and relevance of robot-assisted learning

In section 2, the personality traits were analysed using visual and non-visual data and theoretical distributions were assigned to each score distribution for each personality trait. In this section the effects and relevance of robot-assisted learning for this report will be analysed in different ways using the personality traits. In section 3.1 the potential effects of robot-assisted learning aswell as the potential effects of training on the patients will be numerically analysed and a conclusion will be drawn on how much robot-assisted learning contributes to the reduction of the error. In section 3.2 linear regression models will be used to model the potential effects of robot-assisted learning, and how it affected each personality trait specifically.

3.1 Numerical analysis of effects of robot-assisted learning

To analyse if robot-assisted learning has an effect of error reduction on the patients, the patients get divided into two groups and three categories. The groups consist of the "Control"-group (C) and "Experimental-group". The patients for the control-group does not receive any guidance from the robot while the patients for the experimental group do receive guidance. Patients who are chosen to be in the experimental group are randomly picked out of the 100 patients. The categories are the following: BL, STR and LTR. The patients play the game once to get used to the controls and the game itself, at this point the scores are not logged yet. After that the patients play the game twice, these scores are logged and fall under the stage BL. After the patients played the game twice, they take a short break of 10 minutes, after the break they will play the game twice again, this time the scores are logged too and these fall under the stage STR. After a few days, the patients play the game twice again, these scores are logged again and fall under the stage LTR.

For each stage, there are different errors measured per patient. To model this compactly, the Average Absolute Error of patient i in a certain stage is defined as follows:

$$\text{AveAbsError}_{i,Stage} := \tfrac{1}{40} \textstyle\sum_{k=1}^{20} |\text{Error}|_{i,Stage,1,k} + |\text{Error}_{i,Stage,2,k}|. \quad (3)$$

This represent the absolute Error of patient i from a certain stage in game 1 added to the absolute Error of patient i in game 2 over all 20 targets for both games.

The variance of the absolute errors for each stage and each group can be estimated by the sample variance. The sample variance is defined as:

$S = \frac{1}{n-1} \sum_{i=1}^{n}(X_i - \bar{X})$, where n denotes the size of the group or stage. Since the variance isnt known and is estimated by the sample variance, a paired t-test is an appropriate test statistic to determine whether there has been a significant difference in AveAbsError between groups and/or stages (Since there is no evidence that the data is not normally distributed with p-value¿0.05). If it is proven that there is indeed a difference, a conclusion can be drawn from the effectiveness of robot-assisted learning aswell as how the training helps the patients perform.

A t-test was performed on group C and group E of the dataset, to test whether the final results between the control and experimental group were different, this difference may have come from either robot assistance, training or inherent competence of the patients. In this case the p-values were used again as a measure to determine whether the difference was statistically significant or not, the critical value was defined if p-value< 0.05. The results are as follows:

Stage	Groups compared	p-value	Significance
BL	Control and Experimental	0.2739	Not significant
STR	Control and Experimental	$\sim 10^{-4}$	Significant
LTR	Control and Experimental	$\sim 10^{-4}$	Significant

Table 2: Significance of possible effects for each stage

The results show that for the baseline stage, the final results for the average of the average absolute errors between the control group and the experimental group did not differ a lot from eachother, since there is no training nor robot-assistance at the stage BL, this indicates that the average over the inherent competences of the patients in the control group did not differ significantly from the competences of the patients in the experimental group. However for the short term- and long term retention stages, the average of the average absolute errors do indeed differ since the difference is statistically significant, this may have come from either training or robot-assistance since these factors do have an influence at the stages STR and LTR.

In the previous analysis, both training and robot-assistance may have played a role in the significant error-reduction for the final result in the stages STR and LTR. A paired t-test was performed to test whether training helped, in this case, the same patients are analysed so a paired t-test is appropriate for this case. The results are the following:

Stages compared	p-value	Significance
BL and STR	$\sim 10^{-72}$	Significant
BL and LTR	$\sim 10^{-59}$	Significant

Table 3: Significance of training for STR and LTR stages

By the very low p-values, it can be concluded that training made a significant difference in both stages in the error reduction of patients.

The following measures were used in order to model if robot-assistance only made a significant difference in the error reductions:

$\text{LearningEffect}_{i,STR} := \text{AveAbsError}_{i,BL} - \text{AveAbsError}_{i,STR}$ and

$\text{LearningEffect}_{i,LTR} := \text{AveAbsError}_{i,BL} - \text{AveAbsError}_{i,LTR}$. (4)

This measure was used in order to clean out any advantage Group C had at the baseline stage in comparison to Group E at the baseline stage, Group C for example could have started with an AveAbsError of 10 while Group E could have started with an AveAbsError of 10.5. Even if this difference is not significant at Baseline level (p-value of 0.27 and in this case $10.5 - 10 = 0.5$), this might still have an effect on comparing if only robot-assistant learning had an effect. This measure was not needed when comparing for training effects as then, the same patients at different stages were compared, however now it is needed since different patients in different groups are compared at the same stage. In order to model if the average of the average absolute errors of the control group differed significantly from the average of the average absolute errors of the experimental group only caused by robot assistance using this measure, a t-test was performed on the Learning effect for stages STR and LTR between the control and experimental group, the results are the following:

Stage	Groups compared	p-value	Significance
STR	Control and Experimental	$\sim 10^{-7}$	Significant
LTR	Control and Experimental	$\sim 10^{-13}$	Significant

Table 4: Significance of robot-assistance for STR and LTR stages

From the results above it can be concluded, that robot-assisted learning and training indeed made a significant difference in the error reduction of patients.

3.2 Analysis using linear regression

To analyse the effects of robot-assisted learning and training using linear regression, six linear regression models for the three phases and two groups were made individually. For the stages $s \in \{\mathrm{BL}, \mathrm{STR}, \mathrm{LTR}\}$ the following linear regression model will be used:

$$\mathrm{AveAbsError}_{i,s} = \beta_{0,s} + \beta_{1,s}\mathrm{AC}_i + \beta_{2,s}\mathrm{FS}_i + \beta_{3,s}\mathrm{TC}_i + \beta_{4,s}\mathrm{TB}_i + \epsilon_{i,s}. \quad (5)$$

Below are the tables given for the estimates of the variables and their significance for each linear regression model for all three stages s:

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	0.6754	0.1143	5.91	< 0.001***
AC	0.0004	0.0006	0.58	0.566
FS	0.0125	0.0017	7.29	< 0.001***
TC	-0.0083	0.0006	-14.90	< 0.001***
TB	0.0003	0.0004	0.82	0.415
R^2		0.782		
Adjusted R^2		0.773		
Residual SE		0.0679		
N		100		

Table 1: Regression variables for personality traits for BL-stage

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	0.2265	0.1120	2.02	0.046*
AC	-2.46×10^{-5}	0.000631	-0.04	0.969
FS	0.0116	0.00168	6.90	< 0.001***
TC	-0.00575	0.000548	-10.50	< 0.001***
TB	0.00122	0.000370	3.31	0.001**
R^2		0.676		
Adj. R^2		0.663		
Residual SE		0.0666		
N		100		

Table 2: Regression variables for personality traits for STR-stage

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	0.3158	0.1308	2.41	0.018*
AC	-0.000673	0.000737	-0.91	0.363
FS	0.0146	0.00196	7.44	< 0.001***
TC	-0.00795	0.000640	-12.43	< 0.001***
TB	0.00171	0.000432	3.96	< 0.001***
R^2		0.722		
Adj. R^2		0.711		
Residual SE		0.0778		
N		100		

Table 3: Regression variables for personality traits for LTR-stage

For the regression of BL it can be seen that the intercept, FS and TC are statistically significant and contribute most to this linear regression model (This can be seen by stars *** next to the variable). FS has a positive coefficient, meaning that higher FS values contribute to higher average absolute errors. The opposite is true for TC as the coefficient of that variable is negative. This suggests that on average for the baseline stage that FS and TC are most significant out of the four personality traits for the contribution to AveAbsError, and that FS, in contrast to TC, has a detrimental effect on AveAbsError. The statistically significant intercept could suggest that the game itself is experienced as difficult by the patients apart from their personality traits. This can be seen from the fact that a patient with a score 0 for every personality trait would still have an average absolute error, which is in this case equal to the intercept.. This makes "intuitive" sense as first of all, patients with the FS personality trait are more prone to performing unusual actions, and patients with the TC personality trait are more prone to being motivated by the difficult game. A possible reason why the game itself might be experienced as difficult is because the baseline stage can be viewed as the beginning stage for the patients.

For the regression of STR it is true that FS, TC and TB are statistically significant while the intercept is less statistically significant. This signifies that the game for the patients has become easier on average, which is logical since the patients are now more familiar with the game itself. The most notable change for this stage's linear regression model is that TB has become statistically significant for this model, in contrast to the linear regression model for baseline stage. This suggests that the training has had a negative effect on patients form TB personality trait for the contribution to AveAbsError since the coefficient of TB is positive.

For the regression of LTR, it can be seen that the personality trait TB became even more statistically significant, this suggests the personality trait TB matters more in the LTR stage after training, its coefficient also increased in comparison to its coefficient in the linear regression model for the stage STR.

Now that the effects of training have been highlighted, the effects of robot-assisted learning will be analysed. For this a more extended linear regression model is defined as follows:

$\text{AveAbsError}_{i,s} = \beta_{0,s} + \beta_{1,s}\text{AC}_i + \beta_{2,s}\text{FS}_i + \beta_{3,s}\text{TC}_i + \beta_{4,s}\text{TB}_i + (\beta_{5,s} + \beta_{6,s}\text{AC}_i + \beta_{7,s}\text{FS}_i + \beta_{8,s}\text{TC}_i + \beta_{9,s}\text{TB}_i)\mathbf{1}_{\{\text{Group}_i = E\}} + \epsilon_{i,s}$. (6)

This model also captures the interactions between the variables if they are in the experimental group or not, this enables analysis to be performed on the difference of the variables where one is not in the experimental group and the other one is and to determine whether robot-assisted learning had an effect on each variable individually aswell as in general. Below are the tables given for this linear regression model for all three stages s:

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	0.5211	0.1713	3.04	0.003**
AC	0.00036	0.00091	0.40	0.693
FS	0.01424	0.00261	5.46	< 0.001***
TC	-0.00782	0.00077	-10.12	< 0.001***
TB	0.00032	0.00055	0.58	0.561
GroupE	0.3091	0.2347	1.32	0.191
AC:GroupE	-0.00041	0.00134	-0.31	0.757
FS:GroupE	-0.00289	0.00352	-0.82	0.414
TC:GroupE	-0.00115	0.00116	-0.99	0.324
TB:GroupE	0.00012	0.00078	0.16	0.875
R^2		0.788		
Adj. R^2		0.767		

Table 4: Extended regression variables for personality traits for BL-stage

From the extended regression of BL it can be seen that again the intercept aswell as only FS and TC from the control group are statistically significant. This aligns with the result of table 1 aswell as the result from section 3.1 that for this stage, the difference of AveAbsError between the control and experimental group was not statistically significant.

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	-0.0745	0.1329	-0.56	0.577
AC	0.00061	0.00070	0.87	0.388
FS	0.01521	0.00203	7.51	< 0.001***
TC	-0.00503	0.00060	-8.39	< 0.001***
TB	0.00097	0.00043	2.27	0.026*
GroupE	0.6545	0.1821	3.60	< 0.001***
AC:GroupE	-0.00193	0.00104	-1.85	0.067.
FS:GroupE	-0.00768	0.00273	-2.81	0.006**
TC:GroupE	-0.00088	0.00090	-0.98	0.331
TB:GroupE	0.00015	0.00061	0.24	0.809
R^2		0.803		
Adj. R^2		0.783		

Table 5: Extended regression variables for personality traits for STR-stage

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	0.0050	0.1510	0.03	0.973
AC	0.00025	0.00080	0.31	0.756
FS	0.01851	0.00230	8.04	< 0.001***
TC	-0.00734	0.00068	-10.78	< 0.001***
TB	0.00111	0.00049	2.29	0.024*
GroupE	0.7065	0.2070	3.41	0.001**
AC:GroupE	-0.00273	0.00118	-2.31	0.023*
FS:GroupE	-0.00845	0.00311	-2.72	0.008**
TC:GroupE	-0.00063	0.00103	-0.61	0.542
TB:GroupE	0.00079	0.00069	1.14	0.258
R^2		0.840		
Adj. R^2		0.824		

Table 6: Extended regression variables for personality traits for LTR-stage

There are a few differences for the extended regression of STR in comparison to the extended regression of BL, the intercept is not statistically significant anymore, signifying that the game itself is not experienced as difficult after training. It is also now true that TB, GroupE and FS:GroupE have become statistically significant, with GroupE and FS:GroupE being really significant. The fact that TB is significant now is in line with the results found from table 2. GroupE having a positive coefficient and being statistically significant implies that without accounting for personality traits, GroupE on average raises the value of AveAbsError in comparison to the control group. This however does not mean that the patients in the experimental group perform worse on average than the patients in the control group, in fact, the opposite is true as shown in section 3.1, the interpretation is that if a patient with no resemblance of any of the personality traits were to undergo guidance, then their average error would increase. This is however a case of extreme extrapolation, as seen in section 2.1, nobody has a personality trait score of 0, especially not all personality traits. What is notable is that for the control group FS had a positive coefficient, but in this linear regression model for the experimental group, it has a negative coefficient, this implies that guidance for the stage STR helps people with the FS personality trait to reduce their error.

From the extended regression of LTR it can be seen that now for the linear regression model for the stage LTR it is true that AC:GroupE is also statistically significant. This means that now after having received training, guidance helped patients with the AC personality trait to reduce errors and to positively contribute to AveAbsError (since the coefficient of AC:GroupE is negative in comparison to the coefficient of AC which is positive and thus detrimental for AveAbsError).

As seen in the normal linear regression models for all stages, it is also true that the extended linear regression models have a decent percentage of explained variance, from these models the conclusion can be made that robot-assisted learning has had a positive effect by reducing AveAbsError on certain personality traits after training, for example FS and AC, this also depended on the amount of training they received and thus the stage the patients were in (BL, LTR or STR). Robot-assisted learning did not have a significant effect while the patients were still in the baseline stage, nor for the personality traits TB and TC in the experimental group.

3.3 Linear model performances

In general, it can be seen that all models, both normal and extended, have a decent explained variance rate (R^2). On average the extended models using the term $(\beta_{5,s} + \beta_{6,s}\mathrm{AC}_i + \beta_{7,s}\mathrm{FS}_i + \beta_{8,s}\mathrm{TC}_i + \beta_{9,s}\mathrm{TB}_i)\mathbf{1}_{\{\mathrm{Group}_i = E\}}$ have a better explained variance rate than the normal models. One cause might be that the extended model uses more variables, which increases the explained variance. However the adjusted R^2 which penalizes more interaction terms is on average also higher for the extended than the normal models. This indicates that the extra interaction term $(\beta_{5,s} + \beta_{6,s}\mathrm{AC}_i + \beta_{7,s}\mathrm{FS}_i + \beta_{8,s}\mathrm{TC}_i + \beta_{9,s}\mathrm{TB}_i)\mathbf{1}_{\{\mathrm{Group}_i = E\}}$ helped improve the performance of models in all stages.

4 Power analysis

To test the statistical relevance of the results found in section 3.1 and 3.2 a power test will be performed. In section 4.1 it is explained what a power test is, in section 4.2 a power test will be performed on this project and the results will be highlighted.

4.1 Explanation of a power analysis

A power analysis is a statistical tool used to calculate the power of a statistical experiment. The power is defined as the probability that the study will correctly reject the null hypothesis when it is in fact false. Power analysis is crucial as it not only ensures that the experiment indeed has a sufficient sample size in order to carry out an experiment whose conclusion is statistically meaningful, but also to not waste resources needed for the experiment.

Statistical power can also be viewed as the complement of the probability of a Type II error occurring, this is when the null hypothesis is not rejected when the null hypothesis is actually false. For example, an experiment that has 80 percent power has a 20 percent probability that a type II error occurs. To be able to conduct a power analysis, firstly a null hypothesis and alternative hypothesis need to be defined, after that an effect size, power level and sample size need to be chosen. The effect size is is defined as the magnitude of the relationship or difference that the study wants to detect, a measure for the effect size is for example Cohen's d, for independent samples, Cohen's d is given by the formula:

$d_{\text{Cohen}} = \frac{\mu_1 - \mu_2}{s_{\text{pooled}}}$ (7),

and for paired tests, Cohen's d is given by the formula:

$d_{\text{Cohen}} = \frac{\mu_1 - \mu_2}{s_d}$ (8),

where μ_1 is the mean of group 1, μ_2 is the mean of group 2, s_{pooled} is the pooled standard deviation and s_d is the standard deviation of the differences between the two samples. The significance level is the probability of a type I error, or rejecting a true null hypothesis. A common threshold for the significance level is 0.05. The sample size is the number of participants in the study that directly impacts power. The power level of the research can be calculated once these values have been obtained. The necessary sample size of a research can vice versa be obtained when the wanted power level is known instead of the sample size.

4.2 Power analysis of the research

To be able to determine how many participants are needed per group for this research such that the conclusions about training and robot-assistance are sufficiently powered, three things are needed as noted in section 4.1: the effect size, the significance level and the wanted power level, which is in this case 80 percent. For this research, the amount of participants will be determined with Cohen's d as the measure for the effect size, 0.05 as the significance level. For Cohen's d three values were chosen, 0.2, 0.5 and 0.8, a Cohen's d of 0.2 indicates that the hypothesis of the a priori power testing is that the effect measured is a small effect, a Cohen's d of 0.5 in the same way indicates a medium effect and a Cohen's d of 0.8 indicates a large effect. (Gignac & Szodorai, 2016)

The results of the amount of participants per group needed for each value of Cohen's d, 0.05 significance level and 80 percent power level are as follows:

Effect Size	Cohen's d	Participants per Group	Total Participants (PpG $\times 2$)
Small	0.2	394	788
Medium	0.5	64	128
Large	0.8	26	52

Table 7: Amount of participants needed per group (Control or Experimental) for each value of Cohen's d, 0.05 significance level and 80 percent power for robot-assistance analysis.

The same power analysis can be used to determine the amount of participants per stage needed in order to draw a sufficiently powered conclusion about the effect of training on the patients, the results of the amount of participants per stage needed for each value of Cohen's d, 0.05 significance level and 80 percent power level are as follows:

Effect Size	Cohen's d	Participants per Stage	Total Participants
Small	0.2	199	199
Medium	0.5	34	34
Large	0.8	15	15

Table 8: Amount of participants needed per stage (BL, STR or LTR) for each value of Cohen's d, 0.05 significance level and 80 percent power for training analysis.

To calculate the a posteriori power of the research, we use the same effect size and significance level together with the current amount of participants in the research, namely 50 per group and 100 patients per stage. The associated Cohen d's for the stages STR and LTR per group can be calculated using formulas (1) and (2) aswell as the data above. The results are the following (The control group is denoted as Group C and the experimental group is denoted as Group E):

Effect Size	Cohen's d	Affected conclusion, Groups and Stages	Power level
Very large	1.071	Robot-assistance, STR, Group C and Group E	$\sim$100 percent
Very large	1.772	Robot-assistance, LTR, Group C and Group E	$\sim$100 percent
Very large	4.984	Training, BL and STR	$\sim$100 percent
Very large	3.629	Training, BL and LTR	$\sim$100 percent

Table 9: Power levels for each group and stage for the conclusions about robot-assistant learning and training.

From table 9 it can be seen that the actual Cohen's d''s are very large in comparison to the Cohen's d's chosen in the a priori power analysis. This signifies that the actual effect of both robot-assistance in the STR and LTR stage, aswell as training in the STR- and LTR- in comparison to the BL-stage both helped reduce errors significantly.

From the results above it can thus be concluded that the conclusions for robot-assistant learning and training are sufficiently powered ($\sim$100 percent), which is higher than the wanted 80 percent. The very large effect sizes show that there is indeed a significant difference between two samples.

5 Dependence between personality traits

Modelling the relation between personality traits and how each personality trait depends on eachother can provide useful insights to what extent and how likely a patient with a certain personality possesses another personality trait. In section 5.1, the dependence between personality traits will be analyzed using a correlation matrix and linear regression. In section 5.2, the dependence between personality traits will be analyzed by using a Bayesian approach to the linear regression model.

5.1 Modelling correlation between personality traits

To be able to test dependence between personality traits, a measure for the dependence is needed, the Pearson correlation-coefficient can be used for this measure as it does not rely on theoretical expectations of the distributions of the data, but rather unbiased estimators for the theoretical moments for example sample means which can be calculated using the data itself. The Pearson correlation-coefficient is defined as follows:

$$r_{X,Y} = \frac{\sum_{i=1}^{n}(x_i - \overline{x}_n)(y_i - \overline{y}_n)}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x}_n)^2 \cdot \sum_{i=1}^{n}(y_i - \overline{y}_n)^2}}. \quad (9)$$

Where x_i and y_i are the data points, in this case the scores per personality trait, and where $\overline{x}_n, \overline{y}_n$ denote the sample means of the data points.

The Pearson correlation-coefficient indicates whether there exists a relation between two personality traits, and if so, whether there is a positive or a negative correlation between the personality traits. A positive correlation between personality traits indicates that if a patient scored high on one trait, then it is likely that they scored on high on the other trait aswell. A negative correlation indicates the opposite.

A correlation matrix can be used in order to compactly visualize the correlations between the personality traits. The correlation matrix is as follows:

	AC	**FS**	**TC**	**TB**
AC	1.000	0.538***	−0.139	−0.024
FS	0.538***	1.000	−0.073	−0.334***
TC	−0.139	−0.073	1.000	−0.032
TB	−0.024	−0.334***	−0.032	1.000

Table 10: Correlation matrix between personality traits

From the correlation matrix it can be seen that there are 2 correlations that are denoted with ***, namely the correlation between AC and FS and the correlation between TB and FS. This indicates that the correlation between the two personality traits is very unlikely to have occurred via pure chance, or in statistical terms, that the p-value of the test $H_0 : r_{X,Y} = 0$ against $H_1 : r_{X,Y} \neq 0$ is smaller than 0.001. The correlation between AC and FS is positive, this signifies that patients who scored high on the AC personality trait also most likely scored high on the FS personality trait and vice versa. The opposite is true between FS and TB as the correlation between those two personality traits is negative.

If it is known that there is indeed a correlation between two personality traits, one can also try to calculate how much a personality trait has an effect on the personality trait. This can be done using linear regression models.

For the correlations between AC and FS and between FS and TB we use the following linear regression models:

$\mathrm{FS}_i = \beta_0 + \mathrm{AC}_i\beta_1 + \epsilon_i$ and $\mathrm{FS}_i = \gamma_0 + \mathrm{TB}_i\gamma_1 + \epsilon_i$. (10)

The results for the two linear regression models are the following:

Predictor	**Estimate**	**Std. Error**	**t-value**	**p-value**	**Significance**
Intercept AC	49.09	2.623	18.716	< 0.001	***
AC	0.213	0.033	6.326	<0.001	***
Intercept TB	69.63	1.283	54.251	< 0.001	***
TB	−0.087	0.025	−3.508	< 0.001	***

Table 11: Results of linear regression models

From the results of the linear regression models it can be seen that a one unit increase in FS predicts a 0.213 unit increase in AC, but a 0.087 unit decrease in TB. This implies that people who are more free-spirited can handle boring situations less well than people who are less free-spirited. The opposite is true for the achiever personality trait, as people who are more free-spirited are often more driven to get better scores.

5.2 Bayesian modelling averaging

For normal linear regression, a linear model in the form of $Y_i = \beta_1 x_i + \beta_2 x_i + ... + \beta_n x_i$ is used. The variable Y_i, to test dependence between personality traits, one can analyze this with the help of a linear regression model in the form of for example for AC_i:

$\mathrm{AC}_i = \beta_0 + \beta_1 \mathrm{FS}_i + \beta_2 \mathrm{TC}_i + \beta_3 \mathrm{TB}_i + \epsilon_i$. (11)

This however is one possible model for AC_i based on the variables $\mathrm{FS}_i, \mathrm{TC}_i$ and TB_i. Based on this model, using Bayes' theorem, the probability that this model is the true data generating model can be calculated. The p-values for the variables aswell as whether or not a variable is statistically significant to the model can be calculated.

If TB_i for example is not found to be statistically significant to this linear regression model, one can omit TB_i and test the new model:

$\mathrm{AC}_i = \beta_0 + \beta_1 \mathrm{FS}_i + \beta_2 \mathrm{TC}_i + \epsilon_i$ (12),

and compute the probability of this model being the true data generating model. In this way, every model gets assigned a weight based on a function of their probability.

According to Bayes' theorem, for a model m it is true that:

$\mathbb{P}(Model_m|Data) = \frac{\mathbb{P}(Data|Model_m) \cdot \mathbb{P}(Model_m)}{\mathbb{P}(Data)}$ (13)

To compare model m and model n, the following ratio (Bayes' factor) is observed:

$BF_{mn} = \frac{\mathbb{P}(D|Model_m)}{\mathbb{P}(D|Model_n)}$ (14)

which gives a measure on how much more likely the data D is under model m in comparison to model n.

For a linear regression model $\mathbb{P}(Data|Model_m)$ represents the marginal the likelihood which is defined as follows:

$\mathbb{P}(D|Model_m) = \int \int \mathbb{P}(D|\beta_m, \sigma_m^2, Model_m) \cdot \mathbb{P}(\beta_m, \sigma^2|Model_m) d\beta_m d\sigma_m^2$ (15)

and where $\mathbb{P}(M_m)$ is the prior probability of model m.

To calculate the integral in the expression given for $\mathbb{P}(D|Model_m)$ is often difficult in the practice. The following expression can be used as an approximation for $\mathbb{P}(D|Model_m)$:

$\mathbb{P}(D|Model_m) \approx \exp(-\frac{1}{2}\text{BIC}_m)$,

$\text{BIC}_m = -2\log(\hat{L}_m) + k_m \log(n)$, (16) (Schwarz, 1978)

where $\hat{L}_m$ is the maximized log-likelihood, k_m is the number of parameters and n denotes the sample size.

For all possible k parameters there are a total of 2^k possible linear regression models. Assuming that the prior distribution for each model has equal probability or $\mathbb{P}(Model_m) = \frac{1}{2^k}$ the following formula is obtained for the posterior probability:

$\mathbb{P}(Model_m|D) = \frac{\exp(-\frac{1}{2}\text{BIC}_m)}{\sum_{i=1}^{2^k}\exp(-\frac{1}{2}\text{BIC}_i)}$. (17)

Bayesian modelling averaging also provides a posterior mean estimator for each variable β_j based on the data, namely:

$\mathbb{E}[\beta_j|Data] = \sum_{i=1}^{2^k}\mathbb{E}[\beta_j|Data, Model_m]\cdot\mathbb{P}(Model_m|Data)$ (18)

where $\mathbb{E}[\beta_j|Data, Model_m] = 0$ if variable j is excluded from model m.

The law of total probability can used to calculate the probability that a model variable β_j belongs in the true model:

Denote $I_j = \begin{cases} 1, \beta_j \in Model_m, \\ 0, \text{ otherwise} \end{cases}$ (19)

Now:

$\mathbb{P}(\beta_j \in Model_m) = \mathbb{P}(I_j = 1|Data) = \sum_{i=1}^{2^k}\mathbb{P}(I_j = 1|Model_m, Data)\cdot\mathbb{P}(Model_m|Data) = \sum_{m:\beta_j\in Model_m} 1\cdot\mathbb{P}(Model_m|Data) + \sum_{m:\beta_j\notin Model_m} 0\cdot\mathbb{P}(Model_m|Data) = \sum_{m:\beta_j\in Model_m} P(Model_m|Data)$. (20)

For this research, the same methods that were described were applied, and the results were the following:

Outcome	Variable	PIP	$\boldsymbol{\beta}$ (EV)	Post. Prob.
AC	FS	100%	1.416	$AC \sim FS$ (0.488), $AC \sim FS + TB$ (0.349)
	TC	16.3%	-0.016	$AC \sim FS + TC$ (0.099), $AC \sim FS + TC + TB$ (0.064)
	TB	41.2%	0.047	$AC \sim FS + TB$ (0.349), $AC \sim FS + TC + TB$ (0.064)
TC	FS	8.7%	-0.015	$TC \sim FS$ (0.087)
	AC	17.7%	-0.024	$TC \sim AC$ (0.177)
	TB	7.0%	-0.001	$TC \sim TB$ (0.070)
FS	AC	100%	0.210	$FS \sim AC + TB$ (0.908)
	TC	9.2%	-0.0004	$FS \sim AC + TC + TB$ (0.092)
TB	FS	100%	-1.466	$TB \sim FS$ (0.524), $TB \sim FS + AC$ (0.374)
	AC	41.4%	0.138	, $TB \sim FS + AC + TC$ (0.040)
	TC	10.3%	-0.008	$TB \sim FS + TC$ (0.063)

Note: PIP = Posterior Inclusion Probability, Post. Prob=Posterior Model Probability, EV = Expected Value. Model probabilities sum to 1 for each outcome.

Table 12: Bayesian modelling averaging results for predicting each personality traits with other personality traits

From the results above, a few conclusions can be drawn about the dependence between each personality trait. It can be seen that FS strongly predicts achiever with a coefficient of 1.416 and a posterior probability of one hundred percent. This indicates that FS is very likely to be in the true linear regression model for predicting AC. AC on the other hand also has a posterior probability of 100 percent of predicting FS, but with a much lower coefficient, namely 0.210. This indicates that there is a strong dependence between the two personality traits AC and FS, which is in line with the conclusion drawn from section 5.1 about correlation between personality traits, patients who are more free-spirited are also a lot more likely to be achievers, the opposite effect is also true, but to a lesser degree (around $\frac{1.416}{0.210} \approx 6.7\times$ less strong). This indicates that there exists an asymmetric dependence between FS and AC, namely that being free-spirited makes one much more achievement-oriented (6.7 times stronger) than being achievement-oriented makes one free-spirited.

From the results it can also be seen that FS strongly predicts TB with a posterior probability of one hundred percent and a coefficient of -1.466. This indicates that there is a strong correlation between FS and TB, which is also in line with the conclusion from section 5.1. A patient who is more free-spirited is thus a lot less likely to turn boring events into personal motivation. The opposite is also true with a posterior probability of one hundred percent, but in a much lesser sense (the coefficient of TB in predicting FS is -0.084). Thus this indicates that also here there exists an asymmetric dependence between FS and TB, namely that being free-spirited makes one draw a lot more motivationfrom boring situations than the opposite effect (17$\times$ stronger).

The results of Bayesian modelling averaging are in line with the method of using a correlation matrix, the same personality traits have a high correlation with one another and the rest of the correlations/predictors have a small probability of actually predicting the personality trait chosen, indicating a small probability for actual correlation. This can be seen for example by the fact that for the personality trait TC, all predictors have a small posterior probability of actually predicting TC in the true model.

6 Discussion

Brainstorming on ideas to make the code easy to write and relevant to the problem as well as debugging the code.

7 Conclusion

In this project, the effects of robot-assisted learning on error reduction in a simple target game among 100 injured patients with diverse personality profiles: AC (Achiever), FS (Freedom of Spirit), TC (Transformation of Challenge), and TB (Transformation of Boredom) were examined. Using descriptive plots, distribution fitting, hypothesis tests, linear regression, power analysis, and correlation assessments, the research question: "Can robot-assisted learning help patients with different personality traits reduce errors in a game? was adressed" Key findings indicate that personality traits shape baseline performance, with FS linked to higher errors and TC to lower ones. Training significantly reduces errors from baseline (BL) to short-term retention (STR) and long-term retention (LTR), as confirmed by paired t-tests with p-values $< 10^{-50}$ and large effect sizes. Robot-assisted learning amplifies these gains, with no group differences at BL but substantial error reductions at STR and LTR in the experimental group. Extended regressions highlight trait-specific interactions: guidance benefits FS in STR and both FS and AC in LTR by countering their error-increasing tendencies. Power analysis affirms the results' robustness, yielding around one hundred percent power for detecting these large effects. Trait correlations—positive (AC-FS) and negative (FS-TB)—reveal interdependencies that could guide personalized interventions. Bayesian model averaging reinforces the relevance of these predictors. Ultimately, robot-assisted learning effectively supports error reduction, especially when aligned with personality profiles, with strong potential for rehabilitation in scenarios like post-injury therapy. Future studies might incorporate larger cohorts, more traits, or long-term tracking to enhance these applications.

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Appendix

```
install.packages("carData")
install.packages("car")
install.packages("MASS")
install.packages("fitdistrplus")
install.packages("pwr")
install.packages("BMA")
install.packages("Hmisc")
install.packages("corrplot")
install.packages("xtable")
library("carData")
library("MASS")
library("fitdistrplus")
library("car")
library("pwr")
library("BMA")
library("Hmisc")
library("corrplot")
library("xtable")
d<-read.csv("dataset.csv", header=TRUE)


par(mfrow= c(2,2))
unique_personality <- aggregate(. ~ ParticipantID,
                                data = d[, c("ParticipantID", "Achiever", "FreeSpirit",
                                                "TransformOfChallenge", "TransformOfBorendom")],
                                FUN = function(x) x[1])


par(mfrow = c(2,2))
for(i in 1:4) {
  trait <- c("Achiever", "FreeSpirit", "TransformOfChallenge", "TransformOfBorendom")[i]
  colors <- c("yellow", "green", "blue", "coral")[i]

  hist(unique_personality[[trait]],
        col = colors,
        main = paste(trait, "Score"),
        xlab = paste(trait, "Score␣(0-100)"),
        ylab = "Number␣of␣Participants",
        breaks = 8,
        xlim = c(0, 100))
}
par(mfrow = c(2,2))

FS_unique <- unique_personality$FreeSpirit
AC_unique <- unique_personality$Achiever
TC_unique <- unique_personality$TransformOfChallenge
TB_unique <- unique_personality$TransformOfBorendom

FreeSpirit_normalized <- (FS_unique - min(FS_unique)) / (max(FS_unique) - min(FS_unique))
beta1 <- fitdist(FreeSpirit_normalized, "beta")
qqPlot(FreeSpirit_normalized, distribution = "beta",
        shape1 = beta1$estimate["shape1"],
        shape2 = beta1$estimate["shape2"],
        main = "Beta␣Q-Q␣Plot␣of␣FS␣")
print(beta1)

normal<- fitdist(TB_unique, "norm")
qqPlot(TB_unique, distribution = "norm",
        main = "Normal␣Q-Q␣Plot␣of␣TB␣")
print(normal)

AC_normalized <- (AC_unique-min(AC_unique)) / (max(AC_unique)) / (max(AC_unique))
weibull1<- fitdist(AC_unique, "weibull")
normal1<-fitdist(AC_unique, "norm")
qqPlot(AC_unique, distribution = "weibull",
        shape = weibull1$estimate["shape"],
        scale = weibull1$estimate["scale"],
        main = "Weibull␣Q-Q␣Plot␣of␣AC␣")
print(weibull1)

TransformOfChallenge_normalized <- (TC_unique - min(TC_unique)) / (max(TC_unique) - min(TC_unique))
beta2 <- fitdist(TransformOfChallenge_normalized, "beta")
qqPlot(TransformOfChallenge_normalized, distribution = "beta",
        shape1 = beta2$estimate["shape1"],
        shape2 = beta2$estimate["shape2"],
        main = "Beta␣Q-Q␣Plot␣of␣TC␣")
print(beta2)

ks_fs <- ks.test(FreeSpirit_normalized, "pbeta",
                   shape1 = beta1$estimate["shape1"],
                   shape2 = beta1$estimate["shape2"])
cat("\nKolmogorov-Smirnov␣test␣for␣FreeSpirit␣(Unique␣Values):\n")
print(ks_fs)

ks_tb <- ks.test(TB_unique, "pnorm",
                   mean = normal$estimate["mean"],
                   sd = normal$estimate["sd"])
cat("Kolmogorov-Smirnov␣test␣for␣TransformOfBorendom␣(normal,␣Unique␣Values):\n")
```

```
print(ks_tb)
weibull1<-fitdist(AC_unique, "weibull")
ks_ac <- ks.test(AC_unique, "pweibull",
                 shape = weibull1$estimate["shape"],
                 scale = weibull1$estimate["scale"])
cat("Kolmogorov-Smirnov␣test␣for␣Achiever␣(beta,␣Unique␣Values):\n")
print(ks_ac)

ks_tc <- ks.test(TransformOfChallenge_normalized, "pbeta",
                 shape1 = beta2$estimate["shape1"],
                 shape2 = beta2$estimate["shape2"])
cat("Kolmogorov-Smirnov␣test␣for␣TransformOfChallenge␣(beta,␣Unique␣Values):\n")
print(ks_tc)

results <- data.frame()
for(participant in unique(d$ParticipantID)) {
  for(stage in c("BL", "STR", "LTR")) {
    data_subset <- d[d$ParticipantID == participant & d$Stage == stage, ]

    ave_error <- mean(data_subset$AbsError)

    results <- rbind(results, data.frame(
      ParticipantID = participant,
      Group = data_subset$Group[1],
      Stage = stage,
      AveAbsError = ave_error
    ))
  }
}

wide_data <- data.frame()
for(participant in unique(results$ParticipantID)) {
  data_subset <- results[results$ParticipantID == participant, ]
  wide_data <- rbind(wide_data, data.frame(
    ParticipantID = participant,
    Group = data_subset$Group[1],
    BL = data_subset$AveAbsError[data_subset$Stage == "BL"],
    STR = data_subset$AveAbsError[data_subset$Stage == "STR"],
    LTR = data_subset$AveAbsError[data_subset$Stage == "LTR"]
  ))
}
normalBL<-fitdist(wide_data$BL, "norm")
print(normalBL)

cat(mean(wide_data$BL))
cat(mean(wide_data$STR))
cat(mean(wide_data$LTR))
wide_data$Learn_STR <- wide_data$BL - wide_data$STR
wide_data$Learn_LTR <- wide_data$BL - wide_data$LTR
cohen_d_BL_STR <- mean(wide_data$Learn_STR) / sd(wide_data$Learn_STR)
cohen_d_BL_LTR <- mean(wide_data$Learn_LTR) / sd(wide_data$Learn_LTR)
cat(cohen_d_BL_STR)
cat(cohen_d_BL_LTR)

cohens_d_learn <- function(control_learn, exp_learn) {

  n1 <- length(control_learn)
  n2 <- length(exp_learn)

  mean1 <- mean(control_learn)
  mean2 <- mean(exp_learn)

  sd1 <- sd(control_learn)
  sd2 <- sd(exp_learn)

  pooled_sd <- sqrt(((n1 - 1)*sd1^2 + (n2 - 1)*sd2^2) / (n1 + n2 - 2))

  # Cohen's d
  d <- (mean1 - mean2) / pooled_sd

  return(d)
}


d_learn_str <- cohens_d_learn(
  wide_data$Learn_STR[wide_data$Group == "C"],
  wide_data$Learn_STR[wide_data$Group == "E"]
)

d_learn_ltr <- cohens_d_learn(
  wide_data$Learn_LTR[wide_data$Group == "C"],
  wide_data$Learn_LTR[wide_data$Group == "E"]
)
diffs_str <- wide_data$BL - wide_data$STR
d_str <- mean(diffs_str) / sd(diffs_str)

diffs_ltr <- wide_data$BL - wide_data$LTR
d_ltr <- mean(diffs_ltr) / sd(diffs_ltr)

cat("Cohen's␣d␣for␣(Control␣vs␣Experimental):\n")
cat("STR␣Robot-assistance:␣d␣=", abs(round(d_learn_str, 3)), "\n")
```

```
cat("LTR Robot-assistance: d =", abs(round(d_learn_ltr, 3)), "\n")
cat("STR Training: d=", abs(round(d_str,3)),"\n")
cat("LTR Training d=", abs(round(d_ltr,3)), "\n")


cat("Group differences:\n")
cat("BL p-value:", t.test(wide_data$BL[wide_data$Group == "C"], wide_data$BL[wide_data$Group == "E"])$p.value, "\n")
cat("STR p-value:", t.test(wide_data$STR[wide_data$Group == "C"], wide_data$STR[wide_data$Group == "E"])$p.value, "\n")
cat("LTR p-value:", t.test(wide_data$LTR[wide_data$Group == "C"], wide_data$LTR[wide_data$Group == "E"])$p.value, "\n")

cat("\nDid error reduce?\n")
cat("BL vs STR p-value:", t.test(wide_data$BL, wide_data$STR, paired = TRUE)$p.value, "\n")
cat("BL vs LTR p-value:", t.test(wide_data$BL, wide_data$LTR, paired = TRUE)$p.value, "\n")

cat("\nLearning between groups:\n")
cat("STR learning p-value:", t.test(wide_data$Learn_STR[wide_data$Group == "C"], wide_data$Learn_STR[wide_data$Group == "E"])$p.value, "\n")
cat("LTR learning p-value:", t.test(wide_data$Learn_LTR[wide_data$Group == "C"], wide_data$Learn_LTR[wide_data$Group == "E"])$p.value, "\n")

print(mean(wide_data$Learn_STR[wide_data$Group == "C"]))
print(mean(wide_data$Learn_STR[wide_data$Group == "E"]))
print(mean(wide_data$Learn_LTR[wide_data$Group == "C"]))
print(mean(wide_data$Learn_LTR[wide_data$Group == "E"]))
results <- data.frame()
for(participant in unique(d$ParticipantID)) {
  for(stage in c("BL", "STR", "LTR")) {
    data_subset <- d[d$ParticipantID == participant & d$Stage == stage, ]
    ave_error <- mean(data_subset$AbsError)

    AC <- unique(d$Achiever[d$ParticipantID == participant])[1]
    FS <- unique(d$FreeSpirit[d$ParticipantID == participant])[1]
    TC <- unique(d$TransformOfChallenge[d$ParticipantID == participant])[1]
    TB <- unique(d$TransformOfBorendom[d$ParticipantID == participant])[1]
    Group <- unique(d$Group[d$ParticipantID == participant])[1]

    results <- rbind(results, data.frame(
      ParticipantID = participant,
      Group = Group,
      Stage = stage,
      AveAbsError = ave_error,
      AC = AC, FS = FS, TC = TC, TB = TB
    ))
  }
}

for(stage in c("BL", "STR", "LTR")) {
  stage_data <- results[results$Stage == stage, ]
  model1 <- lm(AveAbsError ~ AC + FS + TC + TB, data = stage_data)
  plot(model1)


  stage_data$GroupE <- ifelse(stage_data$Group == "E", 1, 0)
  model2 <- lm(AveAbsError ~ AC + FS + TC + TB +
  GroupE + GroupE:AC + GroupE:FS + GroupE:TC + GroupE:TB,
           data = stage_data)
  plot(model2)
  print(summary(model2))

  model3 <- lm(FS ~ AC,
               data = stage_data)
  plot(model3)
  print(summary(model3))
}

required_sample1 <- pwr.t.test(power = 0.80,
                               d = 0.8,
                               sig.level = 0.05,
                               type = "two.sample")
required_sample2 <- pwr.t.test(power = 0.80,
                                d = 0.8,
                                sig.level = 0.05,
                                type="paired")

cat("Required sample size for 80% power:\n")
cat(ceiling(required_sample1$n), "participants PER GROUP\n for robot-assistance conclusion")
cat("Total:", ceiling(required_sample1$n) * 2, "participants\n")
cat(ceiling(required_sample2$n), "participants PER STAGE\n for training conclusion")
cat("Total:", ceiling(required_sample2$n), "participants\n")



cat("Control participants:", sum(wide_data$Group == "C"), "\n")
cat("Experimental participants:", sum(wide_data$Group == "E"), "\n")

power1 <- pwr.t.test(n = sum(wide_data$Group=="C"),
                          d = d_learn_str,
                          sig.level = 0.05,
```

```
                             type = "two.sample")
power2 <- pwr.t.test(n = sum(wide_data$Group=="C"),
                      d = d_learn_ltr,
                      sig.level = 0.05,
                      type = "two.sample")
power3 <- pwr.t.test(n = 100,
                      d = d_str,
                      sig.level = 0.05,
                      type = "paired")
power4<-pwr.t.test(n = 100,
                    d = d_ltr,
                    sig.level = 0.05,
                    type="paired")

cat("Current power with 50 participants per group for STR:", round(power1$power * 100, 1), "%\n")
cat("Current power with 50 participants per group for LTR:", round(power2$power *100, 1), "%\n")
cat(round(power3$power *100, 1), "%\n")
cat(round(power4$power *100, 1), "%\n")


model4 <- lm(FS ~ AC, data = stage_data)
model5 <- lm(FS ~ TB, data = stage_data)
plot(model4)
print(summary(model4))
plot(model5)
print(summary(model5))



calculate_aic <- function(loglikelihood, num_parameters) {
  aic <- -2 * loglikelihood + 2 * num_parameters
  return(aic)
}


bl_data <- results[results$Stage == "BL", c("AC", "FS", "TC", "TB")]
str_data <- results[results$Stage == "STR", c("AC", "FS", "TC", "TB")]
ltr_data <- results[results$Stage == "LTR", c("AC", "FS", "TC", "TB")]
personality_data <- unique(results[, c("AC", "FS", "TC", "TB")])


create_cor_matrix <- function(data, stage_name) {
  cat(stage_name, "\n")

  traits <- data[, c("AC", "FS", "TC", "TB")]
  cor_test <- rcorr(as.matrix(traits))

  # Print correlations
  print(round(cor_test$r, 3))
  cat("\nP-values:\n")
  print(round(cor_test$P, 3))

  return(list(matrix = cor_test$r, pvalues = cor_test$P))
}

bl_cor <- create_cor_matrix(bl_data, "Baseline")
str_cor <- create_cor_matrix(str_data, "Short-Term")
ltr_cor <- create_cor_matrix(ltr_data, "Long-Term")


bma_ac <- bicreg(
  x = personality_data[, c("FS", "TC", "TB")],
  y = personality_data$AC,
  strict = FALSE,
  OR = 20
)
bma_tc <- bicreg(
  x = personality_data[, c("FS", "AC", "TB")],
  y = personality_data$TC,
  strict = FALSE,
  OR = 20
)

bma_fs <- bicreg(
  x = personality_data[, c("AC", "TC", "TB")],
  y = personality_data$FS,
  strict = FALSE,
  OR = 20
)

bma_tb <- bicreg(
  x = personality_data[, c("FS", "AC", "TC")],
  y = personality_data$TB,
  strict = FALSE,
  OR = 20
)

summary(bma_ac)
summary(bma_tc)
summary(bma_fs)
summary(bma_tb)
```

Listing 1: Complete R analysis code